

General equation of a conic: $r = \frac{L}{1 + e \cos(\theta)}$ where L is a positive constant
 # L determines the size / scale
 # e here is not the irrational number, but instead 'eccentricity'. It is a positive constant that shows which conic we will have:
 $0 \leq e < 1 \Rightarrow$ ellipse
 $e = 1 \Rightarrow$ parabola
 $e > 1 \Rightarrow$ hyperbola

We will be working off of the general conic section for the majority of this section:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

Circle

When $A=C$ and $B=0$, the conic section forms a circle.

Consider $(x,y) \in \mathbb{R}^2$: $(x-a)^2 + (y-b)^2 = r^2$, $a,b,r \in \mathbb{R}$

An alternate form is $x^2 + y^2 - 2(ax+by) + (a^2+b^2-r^2) = 0$

We can also write the parametric equation for a circle:

$$(x,y) = (a + r \cos \theta, b + r \sin \theta)$$

The Ellipse

An ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (including when it is transformed or rotated).

Notice how when $a=b$, the ellipse is reduced to a circle.

We can also write the parametric equation for an ellipse:

$$(x,y) = (a \cos \theta, b \sin \theta)$$

Parabola

The parabola has the Cartesian equation $y^2 = 4ax$ and parametric equation $(x,y) = (at^2, 2at)$.

You might recognise it as $y = ax^2 + bx + c$ but rotated on its side.

Hyperbola

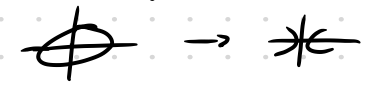
The standard hyperbola has the equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Similarly to the ellipse, it satisfies the parametric equation:

$$(x,y) = (\pm a \cosh \theta, b \sinh \theta)$$

A hyperbola also has asymptotes at $y = \pm \frac{b}{a}x$

Very similar to ellipse (Imagine cutting an ellipse in half, then snapping the sides)



You can show which curve is which by computing $B^2 - 4AC$:

$$\text{Let } \Delta = B^2 - 4AC:$$

$$\text{conic} = \text{ellipse} \Leftrightarrow \Delta < 0$$

$$\text{conic} = \text{parabola} \Leftrightarrow \Delta = 0$$

$$\text{conic} = \text{hyperbola} \Leftrightarrow \Delta > 0$$

Directrix, Focus, and eccentricity

conic sections can be defined using loci.

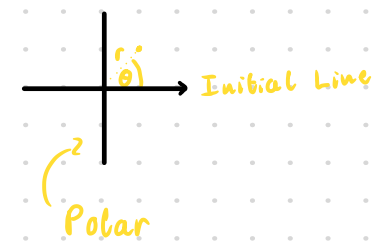
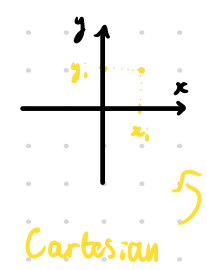
Consider the line D (directrix), the point F (focus), and the eccentricity (e). Given any of these, the set of $P: (x,y)$ such that

$$|PF|^2 = e^2 |PN|^2$$

where the point N is the closest point to P on D .

Polar Curves

We can change coordinate systems from Cartesian to polar.



Note: replacing θ by $\theta + 2\pi$ has no effect on the point

For polar, we take the angle from the initial line (anticlockwise).

The parametric relationship between Cartesian pairs and polar:

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$

e.g. Consider the equation $r = k$ where k is a positive constant

then $r^2 = k^2$, $x = r \cos \theta$ and $y = r \sin \theta$

$$\Rightarrow x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 \text{ or } k^2$$

$\uparrow$ Cartesian equation of circle

Whilst $x = k$ produces a horizontal line, and $y = k$ produces a vertical one, $r = k$ produces a circle and $\theta = k$ produces a ray at the angle θ

Definition: Orthogonal Coordinate Systems

When each family of curves intersect perpendicular to each other, they are considered orthogonal.

Both Cartesian and polar coordinate systems are orthogonal

When plotting, if for some $\theta = \theta^*$ you obtain $r = -r^*$, then instead plot on the ray $\theta = \theta^* + \pi$ with r^* :
 $(-r^*, \theta^*) = (r^*, \theta^* + \pi)$